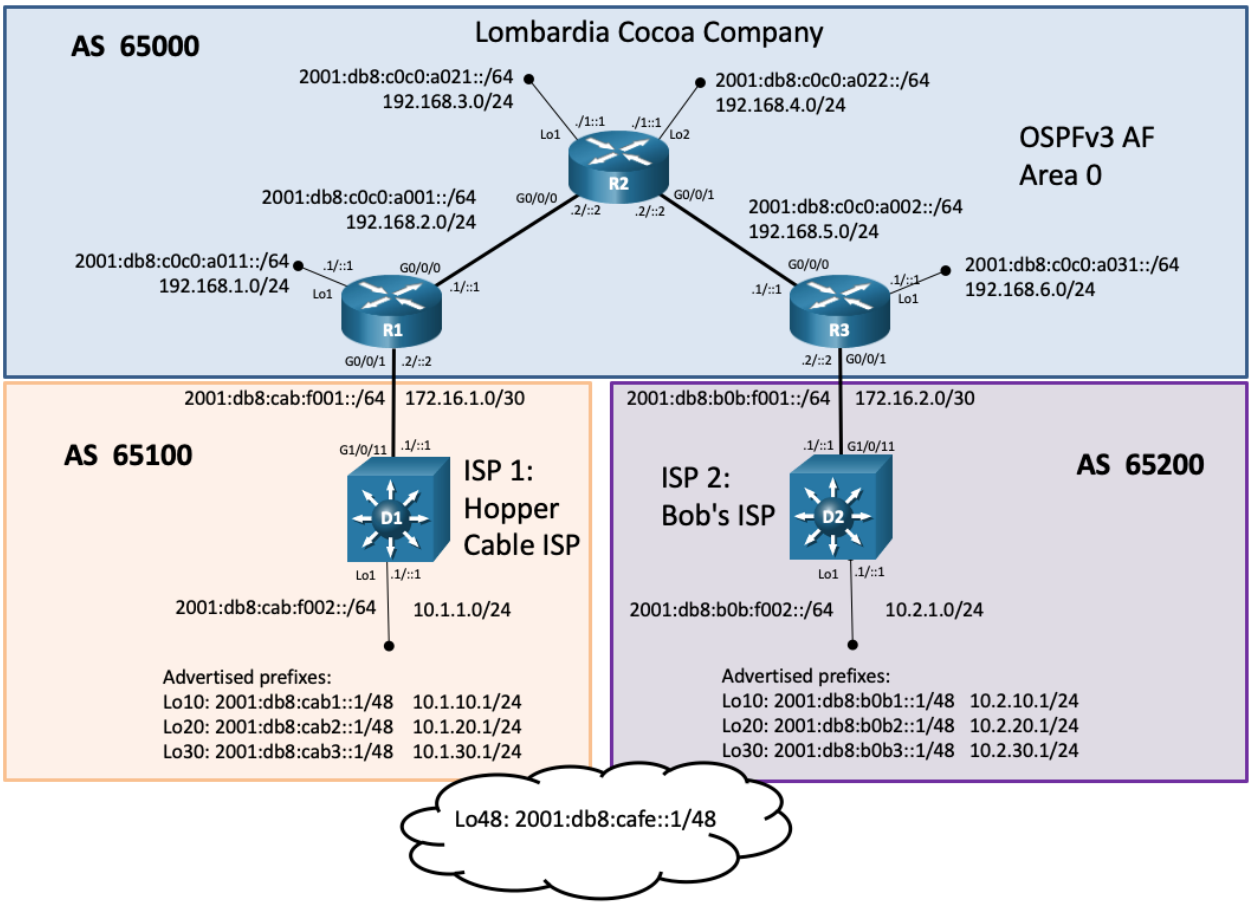
Lab - Troubleshoot BGP

Topology



Note: The advertised prefixes in AS 65100, 65200 and in the "the cloud" are associated with the interface addresses shown (.1 and ::1) for verification purpose. These prefixes are advertised by D1 and D2 to their respective BGP peers.

Note: AS 65100 advertises a shorter AS path to 2001::db8:cafe::/48.

Addressing Table

Device	Interface	IPv6 Address IPv4 Address	IPv6 Link-Local
R1	G0/0/0	2001:db8:c0c0:a001::1/64	fe80::a001:1
		192.168.2.1/24	N/A
	G0/0/1	2001:db8:cab:f001::2/64	fe80::f001:2
		172.16.1.2/30	N/A
	Lo0 (iBGP Peering)	2001:db8:c0c0:99::1/128	fe80::99:1

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Device	Interface	IPv6 Address IPv4 Address	IPv6 Link-Local
		192.168.99.1	N/A
	Lo1	2001:db8:c0c0:a011::1/64	fe80::a011:1
		192.168.1.1/24	N/A
R2	G0/0/0	2001:db8:c0c0:a001::2/64	fe80::a001:2
		192.168.2.2/24	N/A
	G0/0/1	2001:db8:c0c0:a002::2/64	fe80::a002:2
		192.168.5.2/24	N/A
	Lo0 (iBGP Peering)	2001:db8:c0c0:99::2/128	fe80::99:2
		192.168.99.2/24	N/A
	Lo1	2001:db8:c0c0:a021::1/64	fe80::a021:1
		192.168.3.1/24	N/A
	Lo2	2001:db8:c0c0:a022::1/64	fe80::a022:1
		192.168.4.1/24	N/A
R3	G0/0/0	2001:db8:c0c0:a002::1/64	fe80::a002:1
		192.168.5.1/24	N/A
	G0/0/1	2001:db8:b0b:f001::2/64	fe80::f001:2
		172.16.2.2/30	N/A
	Lo0 (iBGP Peering)	2001:db8:c0c0:99::3/128	fe80::99:3
		192.168.99.3/24	N/A
	Lo1	2001:db8:c0c0:a031::1/64	fe80::a031:1
		192.168.6.1/24	N/A
D1	G1/0/11	2001:db8:cab:f001::1/64	fe80::cab:f001:1
		172.16.1.1/30	N/A
	Lo1	2001:db8:cab:f002::1/64	fe80::cab:f002:1
		10.1.1.1/24	N/A
	Lo10	2001:db8:cab1::1/48	EUI-64
		10.1.10.1/24	N/A
	Lo20	2001:db8:cab2::1/48	EUI-64
		10.1.20.1/24	N/A
	Lo30	2001:db8:cab3::1/48	EUI-64
		10.1.30.1/24	N/A
	Lo48	2001:db8:cafe::1/48	EUI-64

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Device	Interface	IPv6 Address IPv4 Address	IPv6 Link-Local
D2	G1/0/11	2001:db8:b0b:f001::1/64	EUI-64
		172.16.2.1/30	N/A
	Lo1	2001:db8:b0b:f002::1/64	EUI-64
		10.2.1.1/24	N/A
	Lo10	2001:db8:b0b1::1/48	EUI-64
		10.2.10.1/24	N/A
	Lo20	2001:db8:b0b2::1/48	EUI-64
		10.2.20.1/24	N/A
	Lo30	2001:db8:b0b3::1/48	EUI-64
		10.2.30.1/24	N/A
	Lo48	2001:db8:cafe::1/48	EUI-64

Note: To make it easier to recognize IPv6 prefixes, familiarize yourself with the IPv6 GUA and LLA address formats.

- GUA: The GUA has a 16-bit subnet-ID, a<area-id><router><network> ("a" for area). For example, subnet-ID a021 is area 0, router 2, network 1.
- LLA: Following best practice the LLA is unique on each interface. The LLA interface-ID uses the GUA subnet-ID:interface-ID for the last 64 bits. For example, fe80::a201:1 has a LLA interface-ID a201 (the subnet-ID of the GUA) and :1 (the interface ID of the GUA).

Objectives

Troubleshoot network issues related to the configuration and operation of MP-BGP with address families.

Background / Scenario

Lombardia Cocoa Company (AS 65000) peers with two ISPs (AS 65100 and AS 65200) and receives the following prefixes from each:

- From ISP1 Hopper Cable ISP (AS 65100):
  - 10.1.1.0/24, 10.1.10.0/24, 10.1.20.0/24, 10.1.30.0/24
  - 2001:db8:cab:f002::/64, 2001:db8:cab1::/48, 2001:db8:cab2::/48, 2001:db8:cab3::/48, 2001:db8:cafe::/48 (shorter AS path)
- From ISP2 Bob's ISP (AS 65200):
  - 10.2.1.0/24, 10.2.10.0/24, 10.2.20.0/24, 10.2.30.0/24
  - 2001:db8:b0b:f002::/64, 2001:db8:b0b2::/48, 2001:db8:b0b2::/48, 2001:db8:b0b3::/48, 2001:db8:cafe::/48 (longer AS path)

Both ISPs also receive specific prefixes from Lombardia.

- Lombardia advertises to both ISPs:
  - 192.168.1.0/24, 192.168.3.0/24, 192.168.4.0/24, 192.168.6.0/24

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- 2001:db8:c0c0:a011::/64, 2001:db8:c0c0:a021::/64, 2001:db8:c0c0:a022::/64, 2001:db8:c0c0:a031::/64

Note: Inter-router links are **not** advertised by BGP.

Note: Lombardia iBGP peering is done using loopback 0 addresses.

Although the topology has a limited number of routers, you should use the appropriate troubleshooting commands to help find and solve the problems in the three trouble tickets as if this were a much more complex topology with many more routers and networks.

You will be loading configurations with intentional errors onto the network. Your tasks are to FIND the error(s), document your findings and the command(s) or method(s) used to fix them, FIX the issue(s) presented here, and then test the network to ensure both of the following conditions are met:

1) the complaint received in the ticket is resolved
2) full reachability is restored

Note: The routers used with CCNP hands-on labs are Cisco 8200L with Cisco IOS XE Release 17.12.5 (universalk9 image). The switches used in the labs are Cisco Catalyst 9300L with Cisco IOS XE Release 17.12.5 (universalk9 image). Other routers, switches, and Cisco IOS versions can be used. Depending on the model and Cisco IOS version, the commands available and the output produced might vary from what is shown in the labs. Refer to the Router Interface Summary Table at the end of the lab for the correct interface identifiers.

Note: Make sure that the devices have been erased and have no startup configurations. If you are unsure, contact your instructor.

Required Resources

- 3 Routers (Cisco 8200L with Cisco IOS XE Release 17.12.5c universal image or comparable)
- 2 Switches (Cisco 9300L with Cisco IOS XE Release 17.12.5 universal image or comparable)
- Console cables to configure the Cisco IOS devices via the console ports
- Ethernet cables as shown in the topology

Part 1: Trouble Ticket 20.1.1.1

Scenario:

Lombardia Cocoa Company is mid-way through a core refresh and resiliency project. As part of the change, router R2 was inserted between the two edge routers (R1 and R3) to provide better path diversity and future scale. The iBGP underlay remains OSPFv3 for IPv4/IPv6, and all iBGP sessions use loopback0 to maintain stable peering. Following the maintenance window, the NOC observed that R1 and R3 only have BGP routes learned from their directly connected ISPs, rather than a complete set of external prefixes across AS 65000. This has led to asymmetric forwarding and intermittent reachability from internal subnets to ISP-learned networks. Your task is to validate the BGP configuration and restore full route visibility so all three AS 65000 routers learn the documented prefixes and traffic no longer relies solely on the directly connected edge.

You have been asked to help find the problem and implement any necessary changes.

Be sure you are using devices that have been connected and prepared by your instructor. Use the commands listed below to load the configuration files for this trouble ticket:

Device	Command
R1	`copy flash:/enarsi/20.1.1.1-r1-config.txt running-config`
R2	`copy flash:/enarsi/20.1.1.1-r2-config.txt running-config`

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Device	Command
R3	**copy flash:/enarsi/20.1.1.1-r3-config.txt running-config**
D1	**copy flash:/enarsi/20.1.1.1-d1-config.txt running-config**
D2	**copy flash:/enarsi/20.1.1.1-d2-config.txt running-config**

- All routers should receive the prefixes shown in the previous Background / Scenario section.
- Verification: The following pings should be successful. If a ping does not work, the ticket is not resolved:

```
R1# ping 10.2.1.1 source lo1
R1# ping 2001:db8:b0b:f002::1 source lo1
R3# ping 10.1.1.1 source lo1
R3# ping 2001:db8:cab:f002::1 source lo1
```

- When you have fixed the ticket, change the MOTD on EACH DEVICE using the following command:

  banner motd # This is $(hostname) FIXED from ticket <ticket number> #

- Then save the configuration by issuing the **wri** command (on each device).
- Inform your instructor that you are ready for the next ticket.
- After the instructor approves your solution for this ticket, issue the **reset.now** privileged EXEC command. This script will clear your configurations and reload the devices.

Part 2: Trouble Ticket 20.1.1.2

Scenario:

Lombardia's finance team has launched a cost-optimization initiative for outbound traffic to the 2001:db8:cafe::/48 prefix. Although Hopper Cable (AS 65100) advertises a shorter AS path for that destination, the company's peering agreements mean traffic exiting via AS 65100 incurs transit charges, while Bob's ISP (AS 65200) does not. A policy update was deployed to prefer AS 65200 for 2001:db8:cafe::/48, but post-change validation shows only R3 is honoring the preference. R1 and R2 continue to select AS 65100, resulting in inconsistent egress and unexpected costs. Your task is to investigate and adjust the BGP policy so the preference for AS 65200 is applied consistently across all AS 65000 routers, with verification showing the best path to 2001:db8:cafe::/48 via AS 65200 on R1, R2, and R3.

You have been tasked to find and resolve the issue.

Use the commands listed below to load the configuration files for this trouble ticket:

Device	Command
R1	**copy flash:/enarsi/20.1.1.2-r1-config.txt running-config**
R2	**copy flash:/enarsi/20.1.1.2-r2-config.txt running-config**
R3	**copy flash:/enarsi/20.1.1.2-r3-config.txt running-config**
D1	**copy flash:/enarsi/20.1.1.2-d1-config.txt running-config**
D2	**copy flash:/enarsi/20.1.1.2-d2-config.txt running-config**

- All AS 65000 routers should forward traffic for 2001:db8:cafe::/48 via AS 65200.
- IPv6 BGP tables in all AS 65000 routers should verify that AS 65200 is the preferred (best) path.
- When you have fixed the ticket, change the MOTD on EACH DEVICE using the following command:

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banner motd # This is $(hostname) FIXED from ticket <ticket number> #

- Then save the configuration by issuing the **wri** command (on each device).
- Inform your instructor that you are ready for the next ticket.
- After the instructor approves your solution for this ticket, issue the **reset.now** privileged EXEC command. This script will clear your configurations and reload the devices.

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